Task 1: Understanding the Business Problem

1.1 Airbnb Market Intelligence Challenge

Business Problem

The business problem investigated in this study is the lack of clear market intelligence about Airbnb accommodation listings, advertised prices, property types, room types, accommodation capacity and customer review activity.

The Airbnb accommodation marketplace contains different types of properties with varying prices, locations, facilities and customer experiences. Understanding these differences is important for identifying market trends, comparing accommodation options and determining which factors may influence advertised prices and customer interest.

This study aims to analyse Airbnb listing and review data to identify patterns and relationships within the accommodation marketplace. Particular attention will be given to differences in advertised prices across room types and locations, the distribution of property categories, accommodation capacity and customer review activity.

Why the Problem Matters to the Accommodation Marketplace

Understanding the Airbnb market is important because accommodation providers operate in a competitive environment where pricing, property characteristics and customer experiences can influence demand and competitiveness.

Market intelligence can help identify pricing differences between accommodation categories, highlight popular property types and locations, and provide insights into customer engagement through review activity.

Without reliable market information, hosts may struggle to position their properties competitively, while accommodation marketplace managers may find it difficult to identify market opportunities and understand customer preferences.

Intended Users of the Solution

The intended users of this market intelligence solution include:

- **Airbnb hosts and property owners:** To understand market pricing, compare similar accommodation listings and improve their property positioning.
- **Airbnb marketplace managers:** To understand listing distribution, market activity and accommodation trends.
- **Potential accommodation investors:** To identify locations and property categories that may offer opportunities in the short-term rental market.

- **Market analysts and researchers:** To investigate pricing patterns, customer review activity and relationships between listing characteristics.

Business Decisions Supported by the Analysis

The findings from this study could support several business decisions, including:

1. **Pricing decisions:** Helping hosts compare advertised prices across room types, property categories and locations.
2. **Property investment decisions:** Identifying property categories and locations that may present potential market opportunities.
3. **Market positioning:** Helping hosts understand how their accommodation compares with similar listings.
4. **Customer experience improvements:** Using review activity and available review information to identify areas requiring further investigation.
5. **Market strategy:** Supporting marketplace managers in understanding listing distribution, accommodation supply and market trends.

Limitations of the Available Evidence

The analysis will be based on the available Airbnb listing and review datasets. However, there are several limitations that must be considered.

- The advertised listing price does not necessarily represent the actual amount paid by guests.
- Listing and review data provide a snapshot of the available information and may not represent current market conditions.
- Review activity does not necessarily represent the overall satisfaction of every guest because not all guests leave reviews.
- Missing values, duplicate records and unusual price values may affect the accuracy of the analysis.
- The available data may not include important factors such as actual booking transactions, occupancy rates, host revenue, operating expenses and seasonal demand.
- Relationships identified through statistical analysis do not necessarily demonstrate cause and effect.

Therefore, the findings should be interpreted as insights into the available Airbnb marketplace data rather than definitive conclusions about actual booking demand, profitability or future market performance.

Conclusion

This study aims to develop a data-driven understanding of the Airbnb accommodation marketplace by analysing listing characteristics, advertised prices, property categories, locations and customer review activity. The resulting insights will support more informed pricing, property positioning, investment research and marketplace strategy decisions.